FINAL PROJECT REPORT

Heart Disease Analysis using Tableau

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1. INTRODUCTION

1.1 Project Overview

Heart disease is one of the leading causes of death worldwide. Many factors such as age, cholesterol levels, blood pressure, smoking habits, and lifestyle choices contribute to the development of heart disease. Healthcare organizations collect large volumes of patient data, but analyzing this data manually can be difficult and time-consuming.

This project focuses on analyzing heart disease data using **Tableau**, a powerful business intelligence and data visualization tool. The system transforms raw healthcare data into interactive visualizations and dashboards that help users understand patterns, correlations, and risk factors associated with heart disease.

By using Tableau dashboards and stories, this project provides meaningful insights that can assist healthcare professionals, researchers, and policymakers in making informed decisions.

1.2 Purpose

The purpose of this project is to analyze heart disease datasets and present insights using interactive dashboards. The project aims to:

- Identify key risk factors associated with heart disease.
- Visualize healthcare data for better understanding.
- Help doctors and healthcare professionals detect high-risk patients.
- Provide decision support for healthcare policy planning.
- Increase awareness of heart disease among individuals.

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2. IDEATION PHASE

2.1 Problem Statement

Healthcare organizations generate large volumes of medical data, but extracting meaningful insights from this data is challenging. Doctors and researchers often struggle to identify patterns and trends in raw datasets.

There is a need for an effective system that can visualize healthcare data and highlight risk factors associated with heart disease. This project aims to address this problem by creating interactive dashboards using Tableau that transform raw healthcare data into understandable visual insights.

2.2 Empathy Map Canvas

User Types

- Doctors and cardiologists
- Healthcare researchers
- Government health departments
- Patients and individuals

User Needs

- Quick analysis of patient data
- Identification of high-risk groups
- Visualization of healthcare patterns
- Easy interpretation of medical data

User Pain Points

- Difficulty analyzing large datasets
- Lack of visualization tools
- Time-consuming manual analysis

User Gains

- Better healthcare decision making
- Improved disease prevention strategies
- Increased awareness of health risks

2.3 Brainstorming

Several ideas were considered during the brainstorming phase, including:

- Building a heart disease prediction model

- Developing a healthcare analytics dashboard
- Creating visual reports for medical data analysis

Finally, the team decided to develop a **Tableau-based Heart Disease Analysis Dashboard** because it provides strong visualization capabilities and allows users to explore data interactively.

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3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. User accesses the system through a web interface.
2. The system retrieves the heart disease dataset from the database.
3. Tableau processes the data and generates visualizations.
4. Users interact with dashboards and filters.
5. Insights are displayed for analysis and decision making.

3.2 Solution Requirement

Functional Requirements

- Import heart disease dataset
- Perform data cleaning and preparation
- Generate visualizations
- Create interactive dashboards
- Provide filters for data exploration
- Integrate dashboards into a web application

Non-Functional Requirements

- User-friendly interface
- Fast data visualization
- Secure data handling
- Scalability for large datasets

3.3 Data Flow Diagram

Insert here:

- **DFD Level 0 Image**
- **Simplified DFD Image**

(Use the diagrams already created.)

The Data Flow Diagram represents how data moves through the system, starting from the dataset to visualization and user interaction.

3.4 Technology Stack

Technology	Purpose
Tableau	Data Visualization and Dashboard Creation
SQL Database	Data Storage and Management
Python Flask	Web Application Integration
Excel / CSV	Dataset Storage
HTML/CSS	User Interface

4. PROJECT DESIGN

4.1 Problem Solution Fit

The project addresses the problem of analyzing complex healthcare datasets by providing interactive visualizations. Tableau dashboards simplify data interpretation and help identify key health indicators related to heart disease.

4.2 Proposed Solution

The proposed solution is a **Heart Disease Analysis System** that uses Tableau dashboards to analyze healthcare data.

The system collects heart disease datasets, stores them in a database, and performs data preparation. Tableau is then used to create visualizations such as charts and graphs that highlight relationships between medical indicators.

These dashboards allow users to analyze heart disease risk factors such as age, cholesterol levels, blood pressure, and lifestyle habits.

4.3 Solution Architecture

Insert here:

Solution Architecture Diagram

Architecture Components:

1. Data Source – Heart Disease Dataset
2. Database – SQL Storage
3. Data Preparation – Data Cleaning
4. Visualization – Tableau Dashboards
5. Web Application – Flask Integration
6. Users – Doctors, Researchers, Individuals

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5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was completed in the following phases:

1. Problem Identification
2. Data Collection
3. Data Cleaning and Preparation
4. Tableau Visualization
5. Dashboard and Story Creation
6. Web Integration using Flask
7. Testing and Evaluation

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6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The system was tested to ensure efficient data visualization and smooth interaction with dashboards.

Testing included:

- Data rendering speed
- Dashboard responsiveness
- Filter functionality
- Visualization accuracy

The system performed efficiently and handled the dataset successfully.

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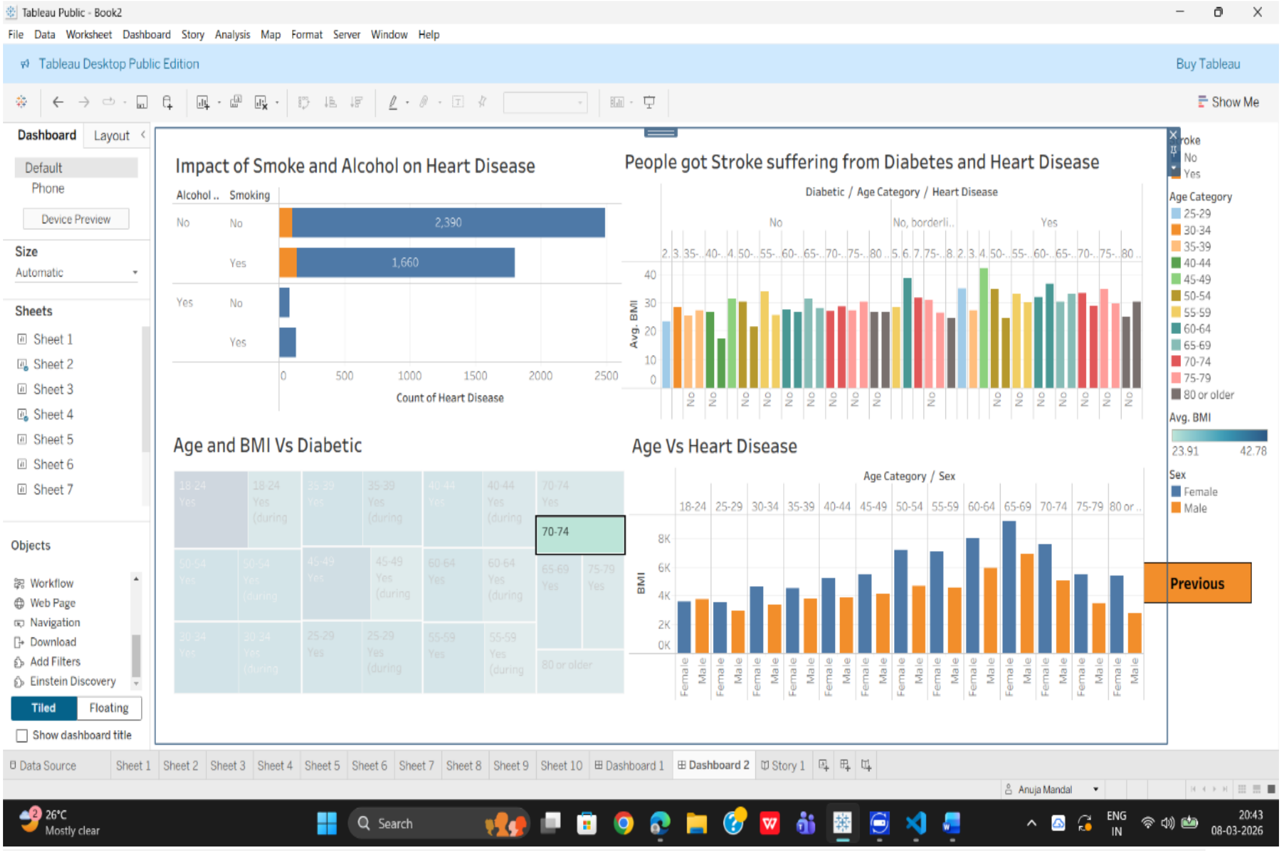
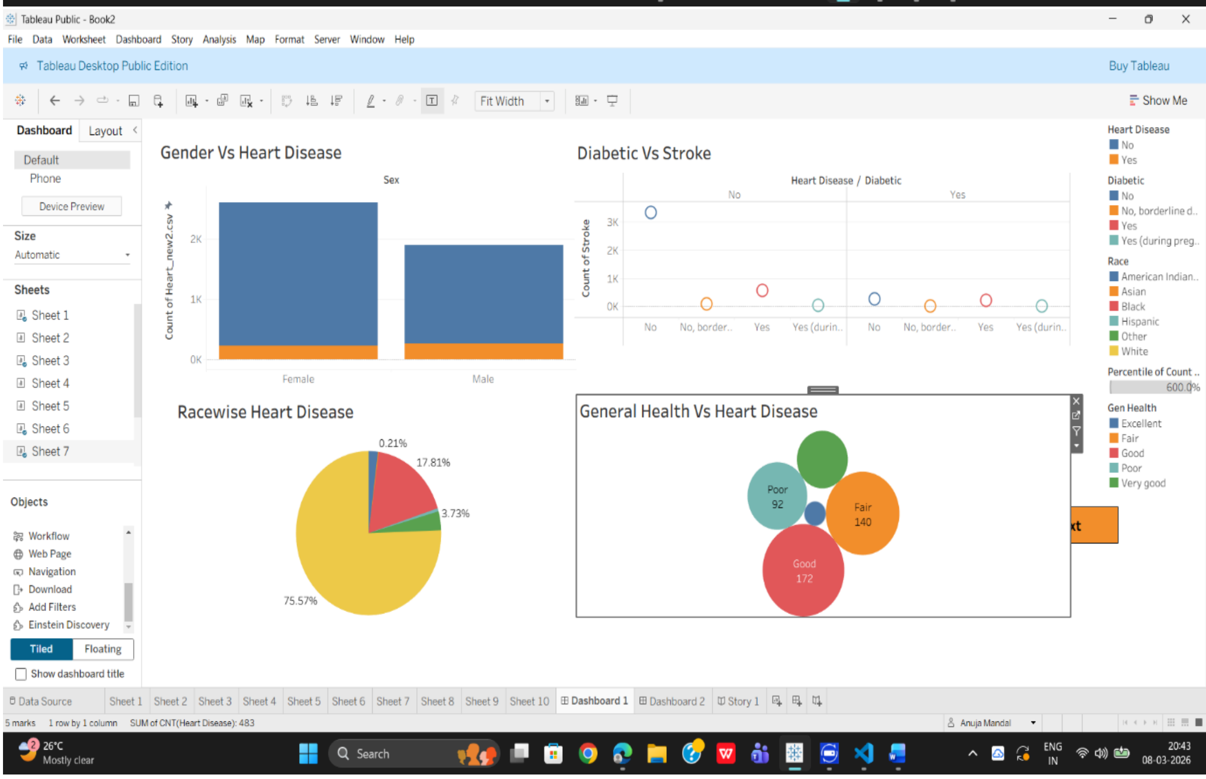
7. RESULTS

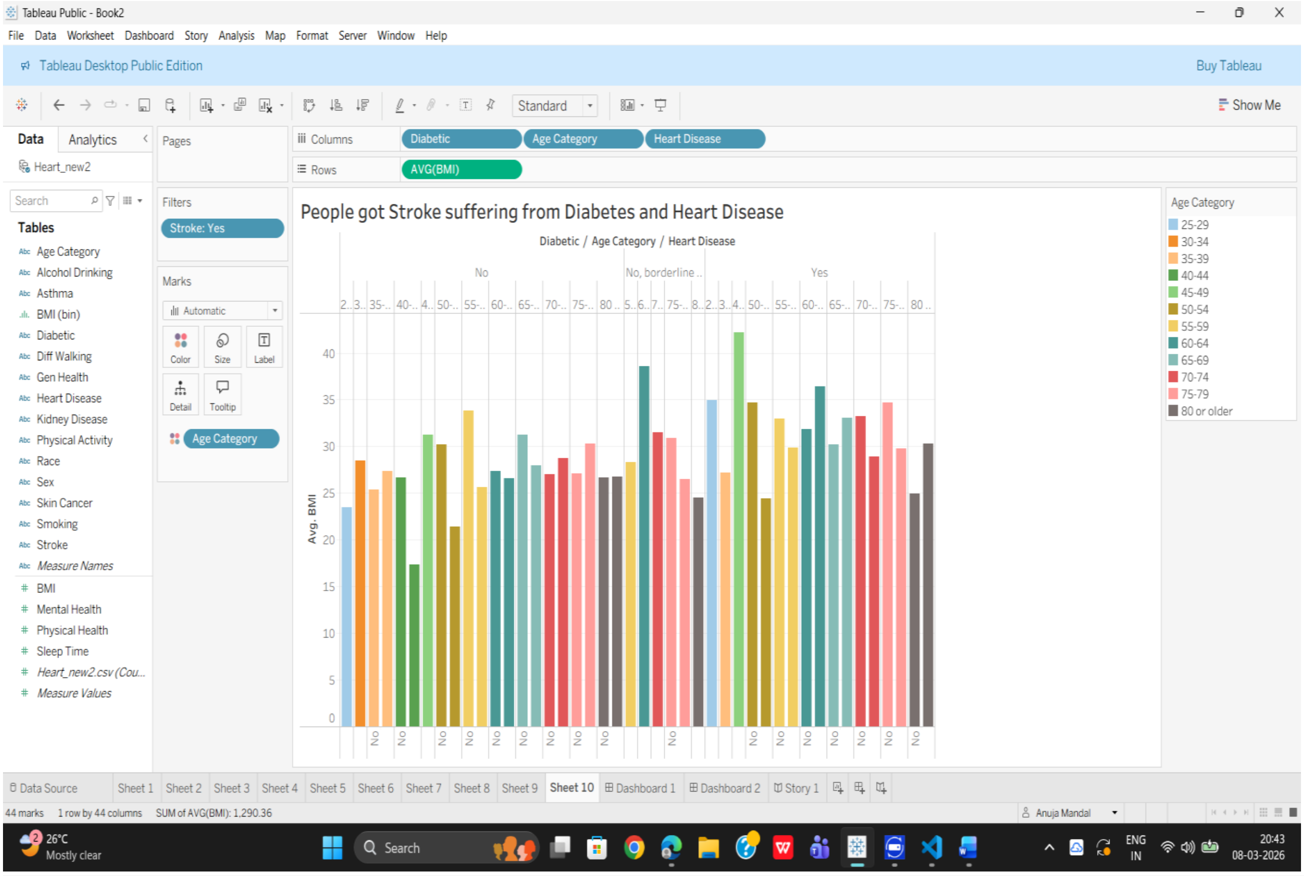
7.1 Output Screenshots

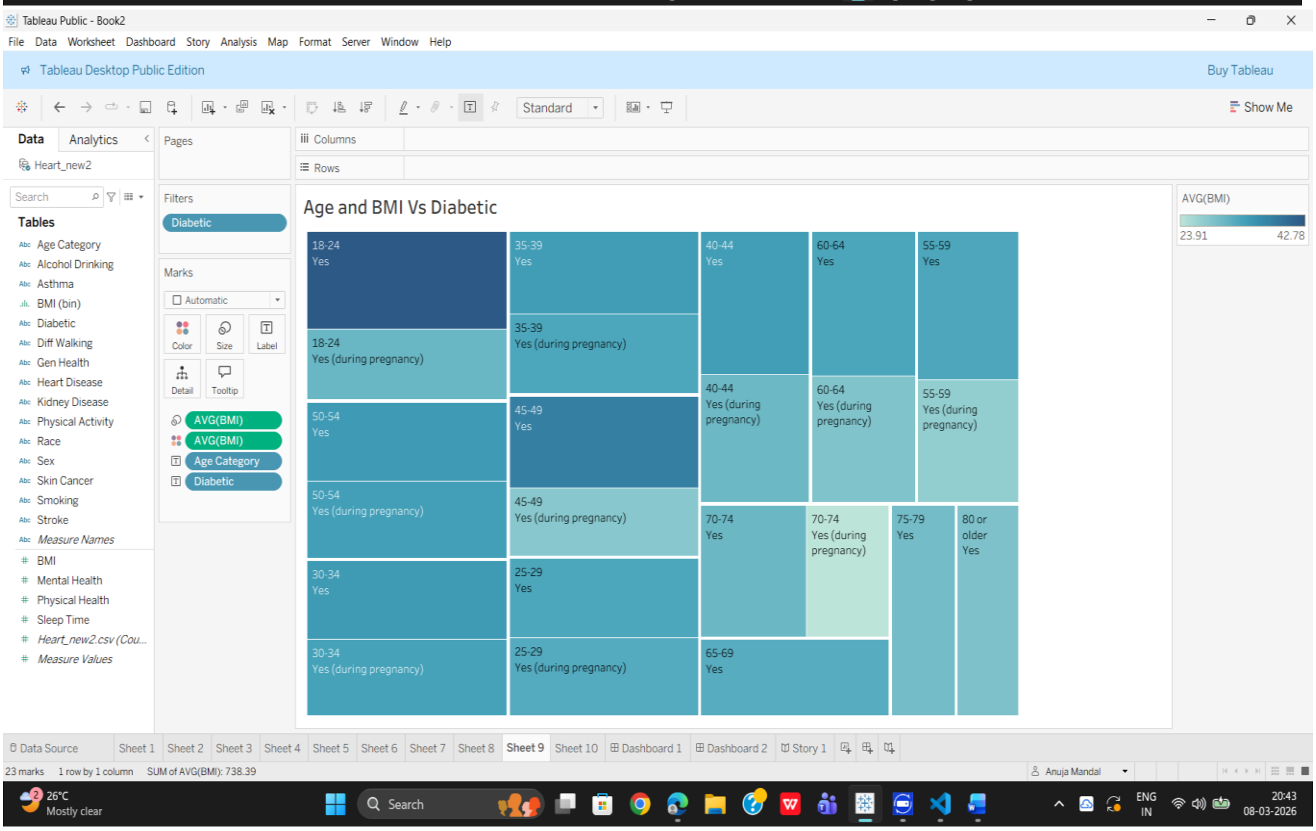
- Tableau Dashboard





- Data Visualization Charts





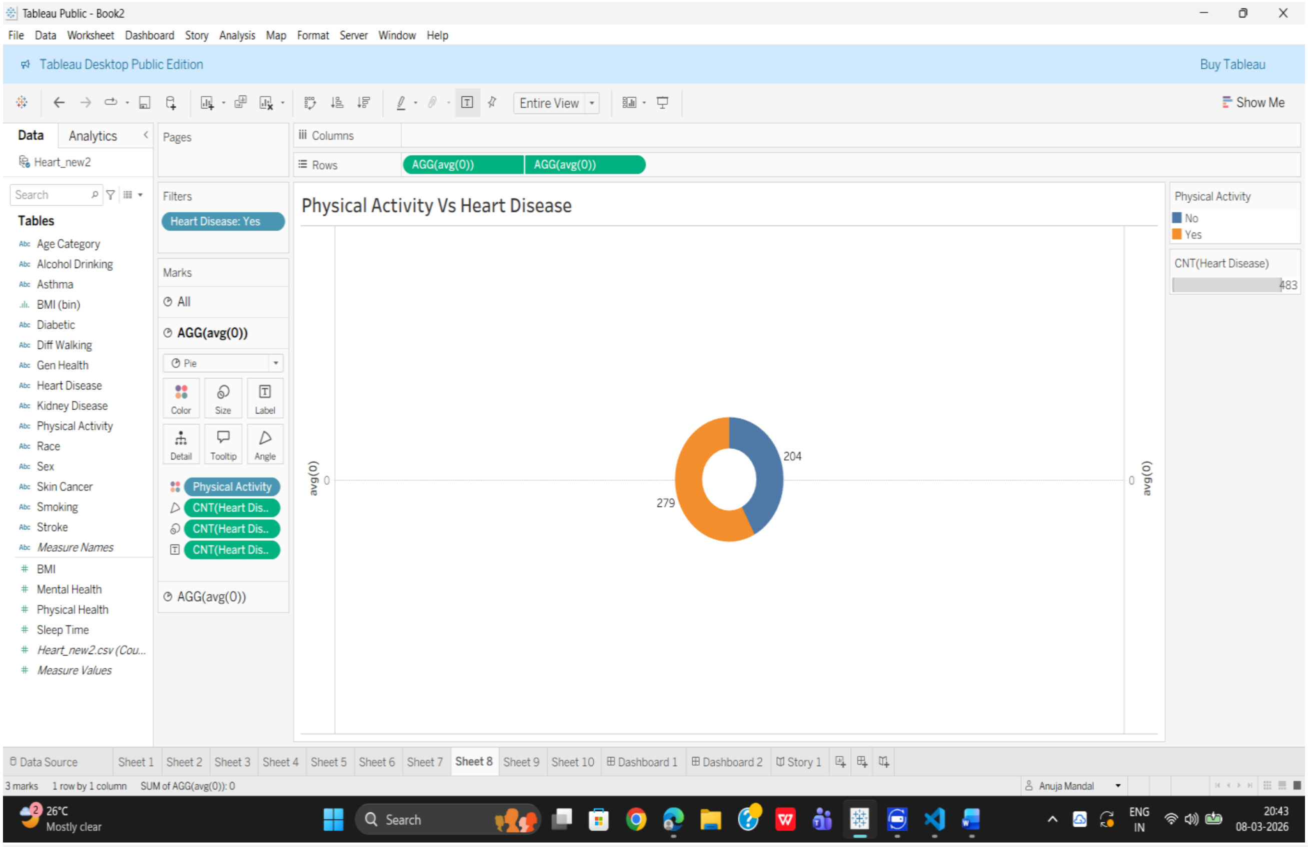


Tableau Public - Book2
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Tableau Desktop Public Edition
Buy Tableau
Entire View
Show Me
Data Analytics
Heart_new2
Pages
Columns
Rows
AGG(avg(0))
AGG(avg(0))
Filters
Heart Disease: Yes
Marks
All
AGG(avg(0))
Pie
Color Size Label
Detail Tooltip Angle
Physical Activity
CNT(Heart Dis..
CNT(Heart Dis..
CNT(Heart Dis..
AGG(avg(0))
Tables
Age Category
Alcohol Drinking
Asthma
BMI (bin)
Diabetic
Diff Walking
Gen Health
Heart Disease
Kidney Disease
Physical Activity
Race
Sex
Skin Cancer
Smoking
Stroke
Measure Names
BMI
Mental Health
Physical Health
Sleep Time
Heart_new2.csv (Cou...
Measure Values
Physical Activity Vs Heart Disease
204
279
avg(0)
0
Physical Activity
No
Yes
CNT(Heart Disease)
483
Data Source Sheet 1 Sheet 2 Sheet 3 Sheet 4 Sheet 5 Sheet 6 Sheet 7 Sheet 8 Sheet 9 Sheet 10 Dashboard 1 Dashboard 2 Story 1
3 marks 1 row by 1 column SUM of AGG(avg(0)): 0
Anuja Mandal
26°C
Mostly clear
Search
ENG
IN
20:43
08-03-2026

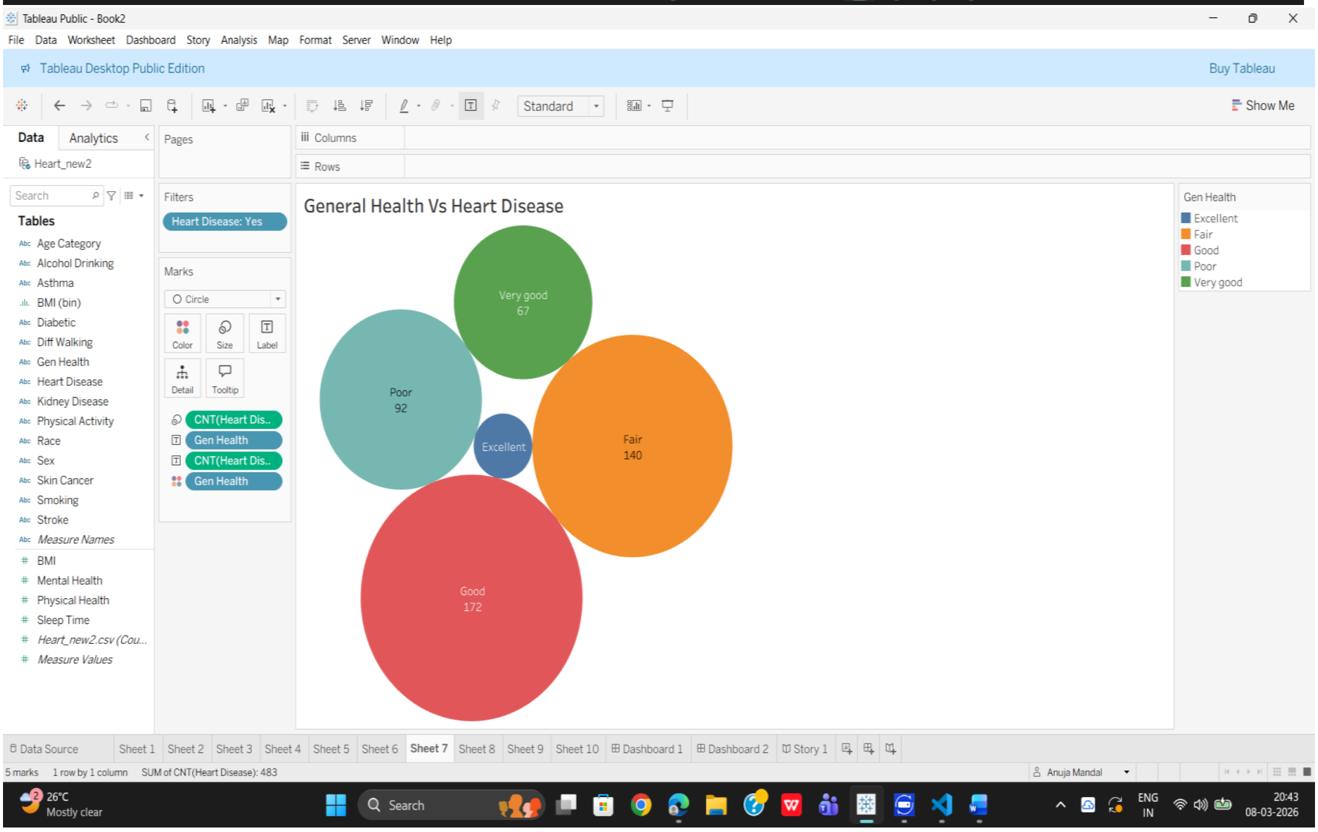


Tableau Public - Book2
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Tableau Desktop Public Edition
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Standard
Show Me
Data Analytics
Heart_new2
Pages
Columns
Rows
Filters
Heart Disease: Yes
Marks
Circle
Color Size Label
Detail Tooltip
CNT(Heart Dis..
Gen Health
CNT(Heart Dis..
Gen Health
Tables
Age Category
Alcohol Drinking
Asthma
BMI (bin)
Diabetic
Diff Walking
Gen Health
Heart Disease
Kidney Disease
Physical Activity
Race
Sex
Skin Cancer
Smoking
Stroke
Measure Names
BMI
Mental Health
Physical Health
Sleep Time
Heart_new2.csv (Cou...
Measure Values
General Health Vs Heart Disease
Very good
67
Poor
92
Excellent
Fair
140
Good
172
Gen Health
Excellent
Fair
Good
Poor
Very good
Data Source Sheet 1 Sheet 2 Sheet 3 Sheet 4 Sheet 5 Sheet 6 Sheet 7 Sheet 8 Sheet 9 Sheet 10 Dashboard 1 Dashboard 2 Story 1
5 marks 1 row by 1 column SUM of CNT(Heart Disease): 483
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26°C
Mostly clear
Search
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IN
20:43
08-03-2026

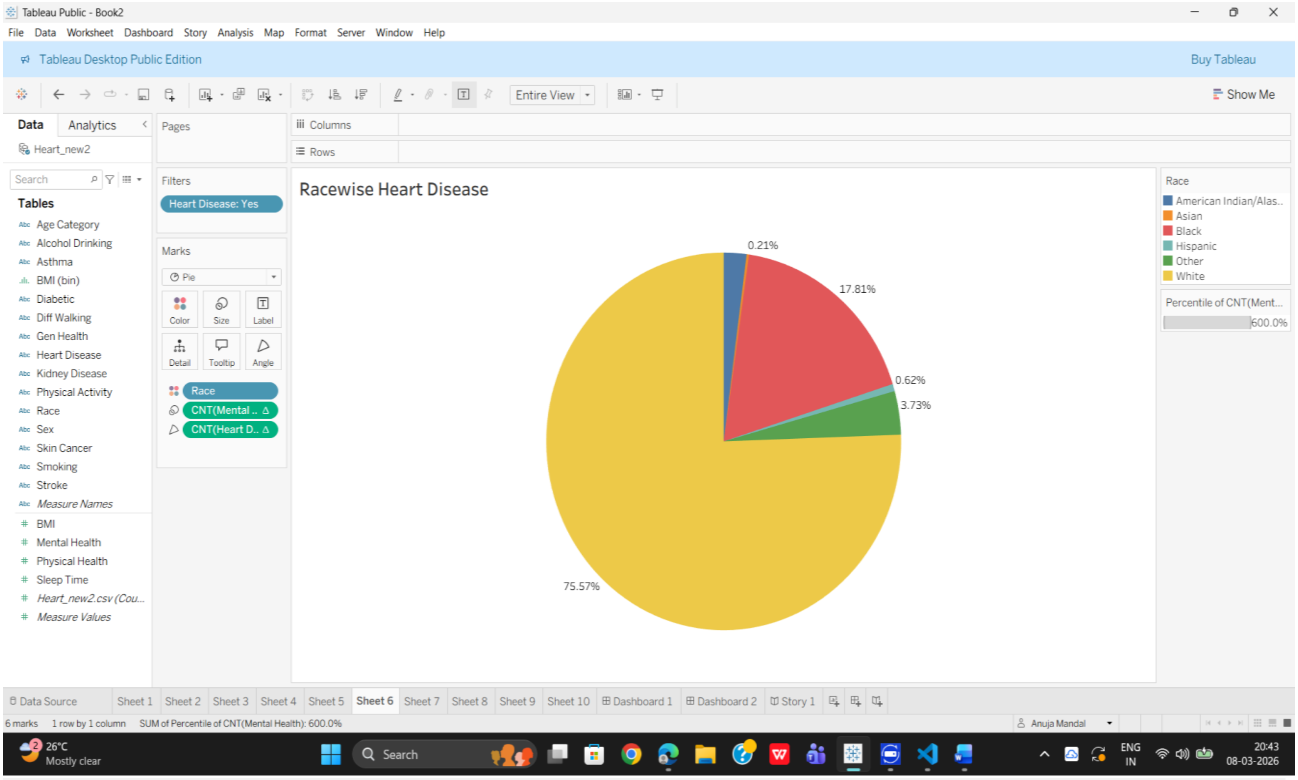
Tableau Public - Book2
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Filters
Heart Disease: Yes
Marks
Pie
Race
CNT(Mental ..
CNT(Heart D..
Racewise Heart Disease
0.21%
17.81%
0.62%
3.73%
75.57%
Race
American Indian/Alas..
Asian
Black
Hispanic
Other
White
Percentile of CNT(Ment...
600.0%
6 marks
1 row by 1 column
SUM of Percentile of CNT(Mental Health): 600.0%

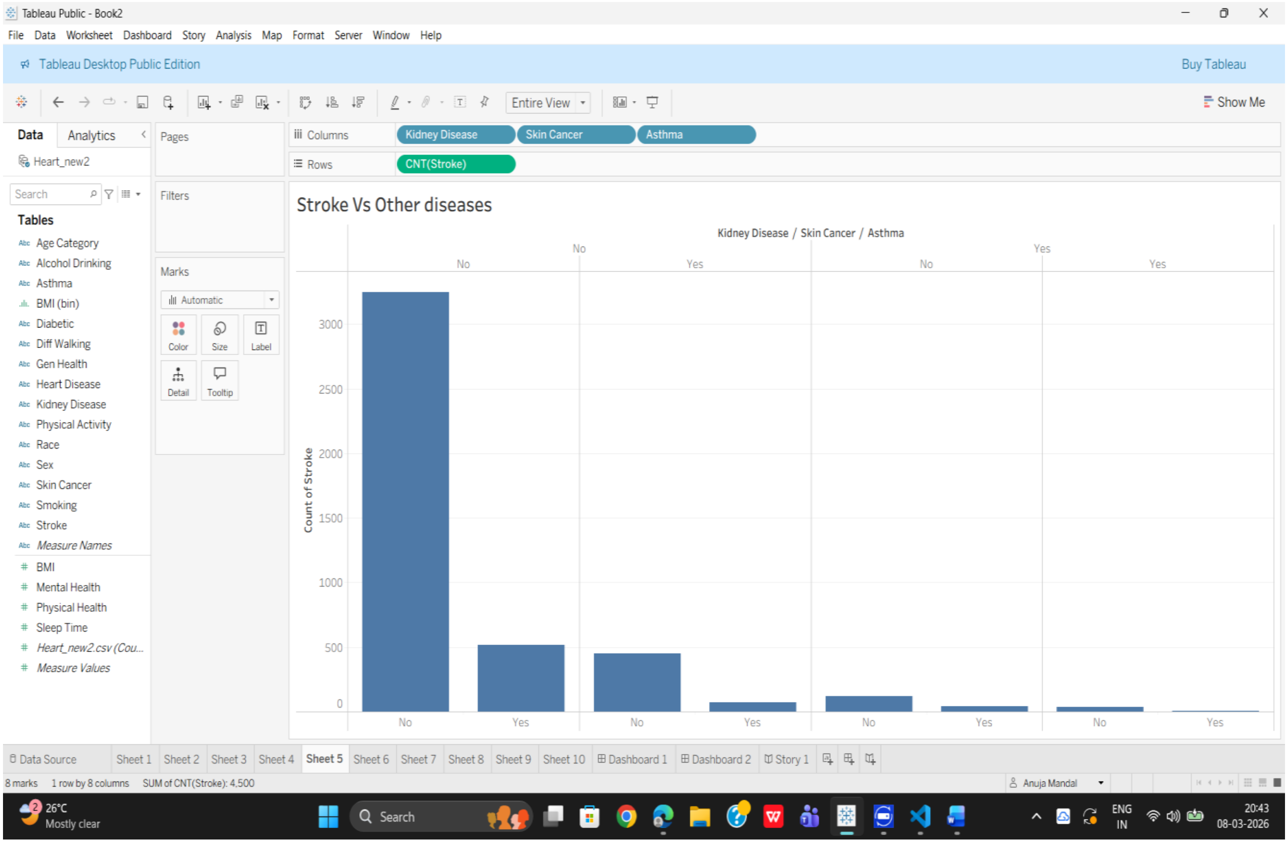
Tableau Public - Book2
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Columns
Kidney Disease
Skin Cancer
Asthma
Rows
CNT(Stroke)
Stroke Vs Other diseases
Kidney Disease / Skin Cancer / Asthma
No
Yes
Count of Stroke
3000
2500
2000
1500
1000
500
0
8 marks
1 row by 8 columns
SUM of CNT(Stroke): 4,500

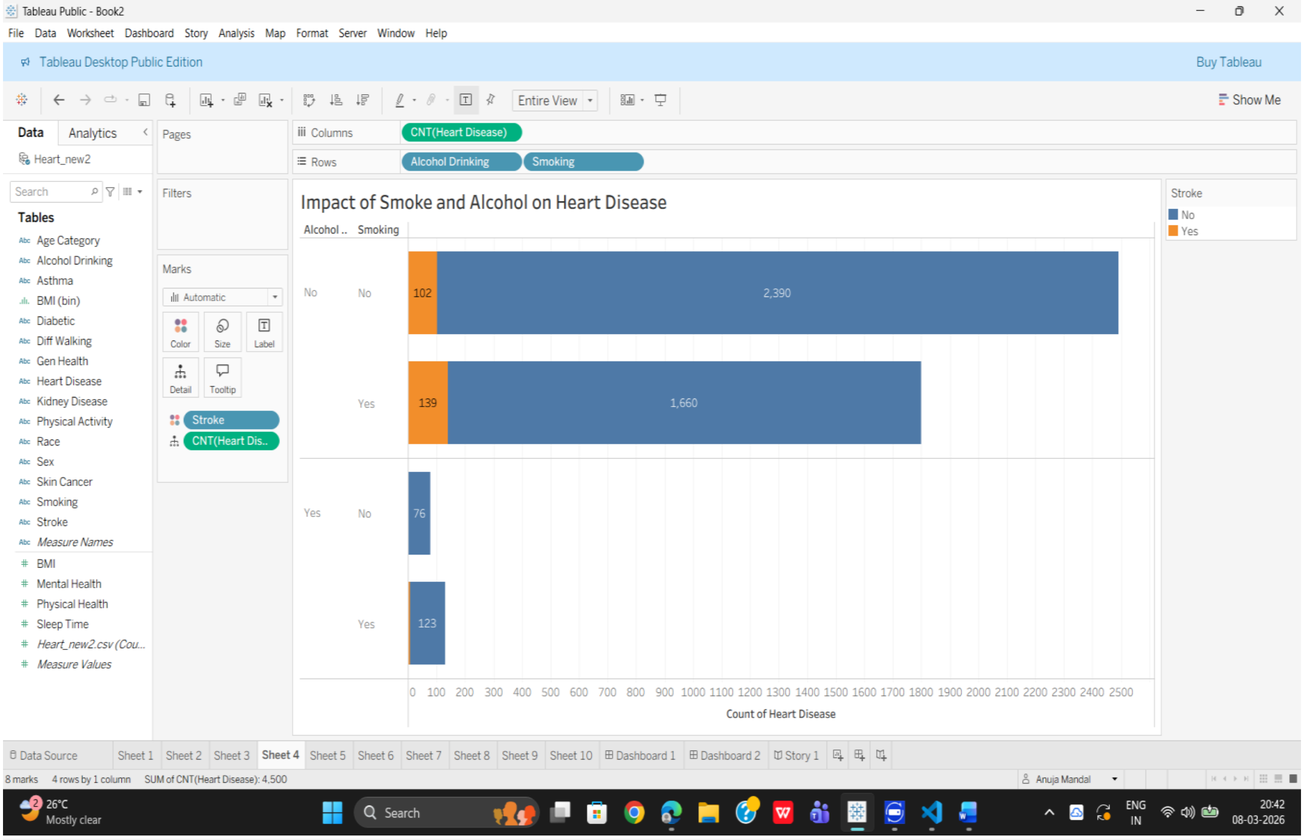
Tableau Public - Book2
Tableau Desktop Public Edition
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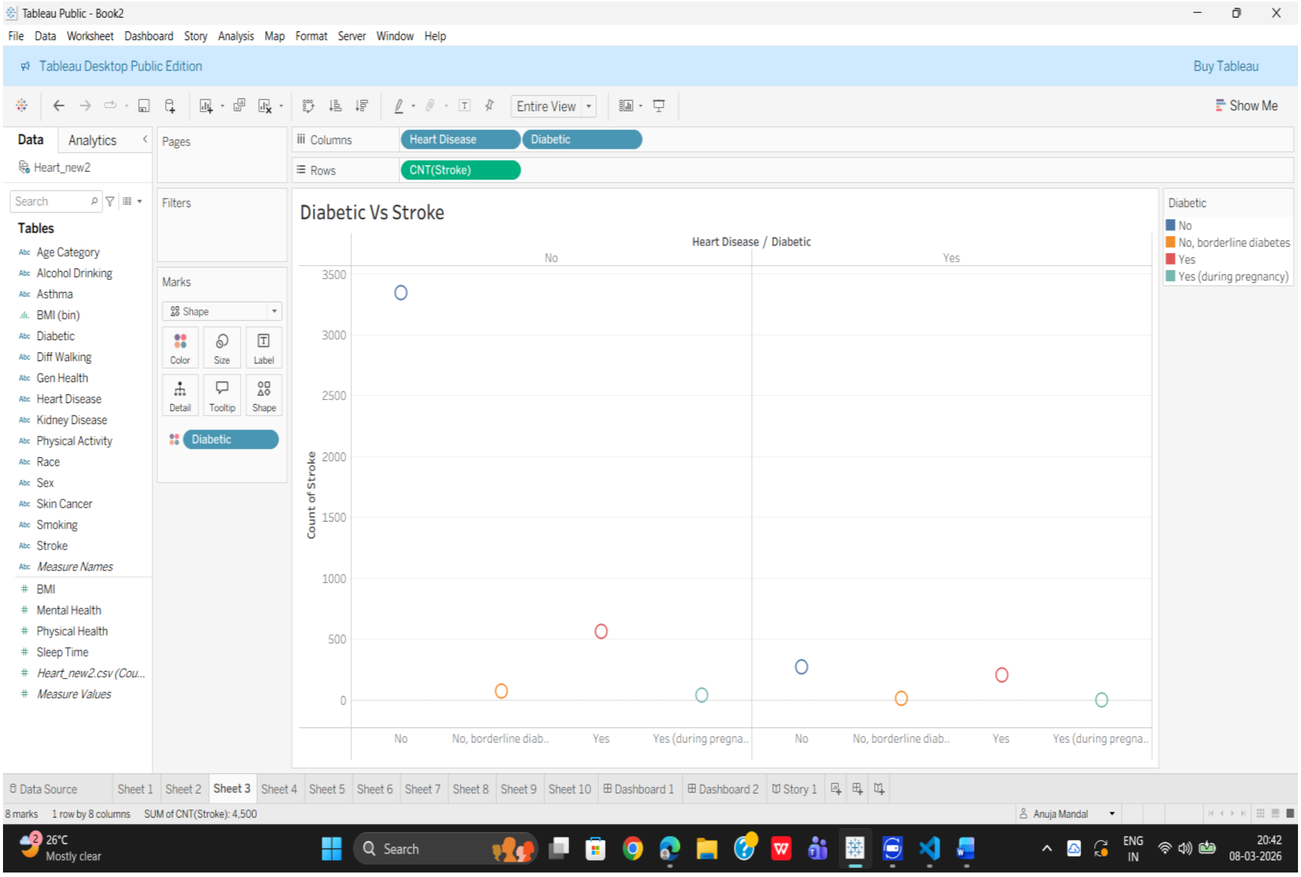
Columns
CNT(Heart Disease)
Rows
Alcohol Drinking
Smoking
Impact of Smoke and Alcohol on Heart Disease
Alcohol ..
Smoking
No
Yes
102
2,390
139
1,660
76
123
0 100 200 300 400 500 600 700 800 900 1000 1100 1200 1300 1400 1500 1600 1700 1800 1900 2000 2100 2200 2300 2400 2500
Count of Heart Disease
Stroke
No
Yes
8 marks 4 rows by 1 column SUM of CNT(Heart Disease): 4,500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Columns
Heart Disease
Diabetic
Rows
CNT(Stroke)
Diabetic Vs Stroke
Heart Disease / Diabetic
No
Yes
Count of Stroke
3500
3000
2500
2000
1500
1000
500
0
No
No, borderline diab..
Yes
Yes (during pregna..
Diabetic
No
No, borderline diabetes
Yes
Yes (during pregnancy)
8 marks 1 row by 8 columns SUM of CNT(Stroke): 4,500

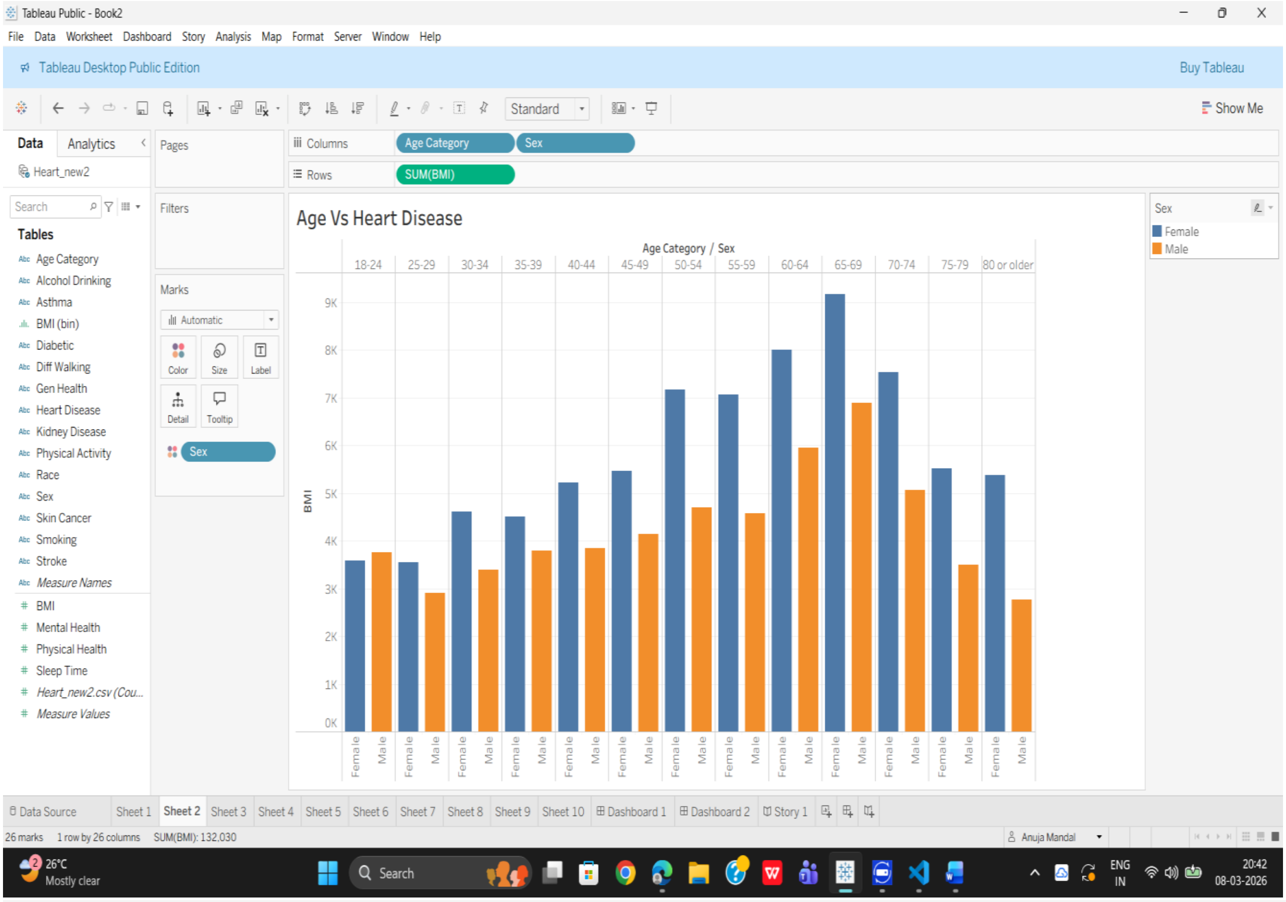


Tableau Public - Book2
Age Vs Heart Disease
Age Category / Sex
18-24 25-29 30-34 35-39 40-44 45-49 50-54 55-59 60-64 65-69 70-74 75-79 80 or older
BMI
Sex
Female
Male
26 marks 1 row by 26 columns SUM(BMI): 132,030

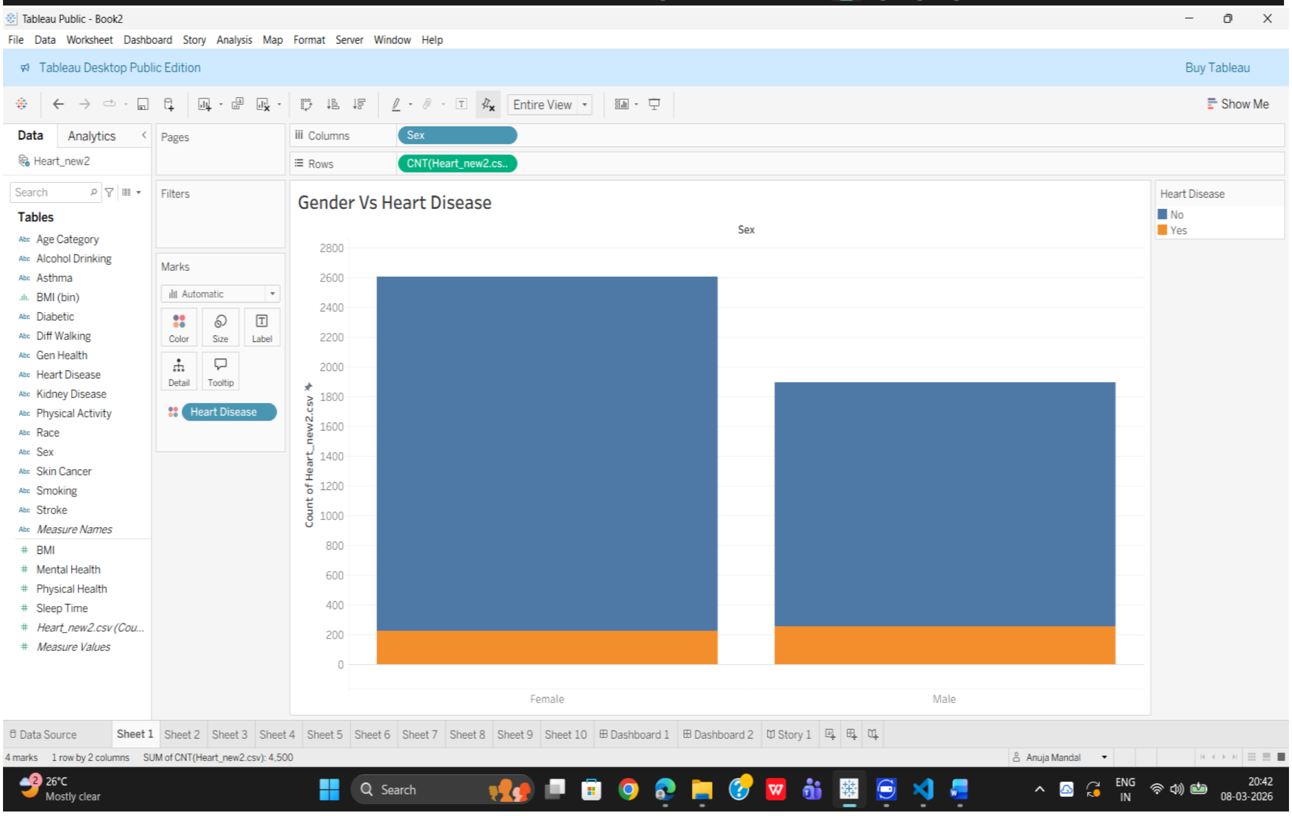
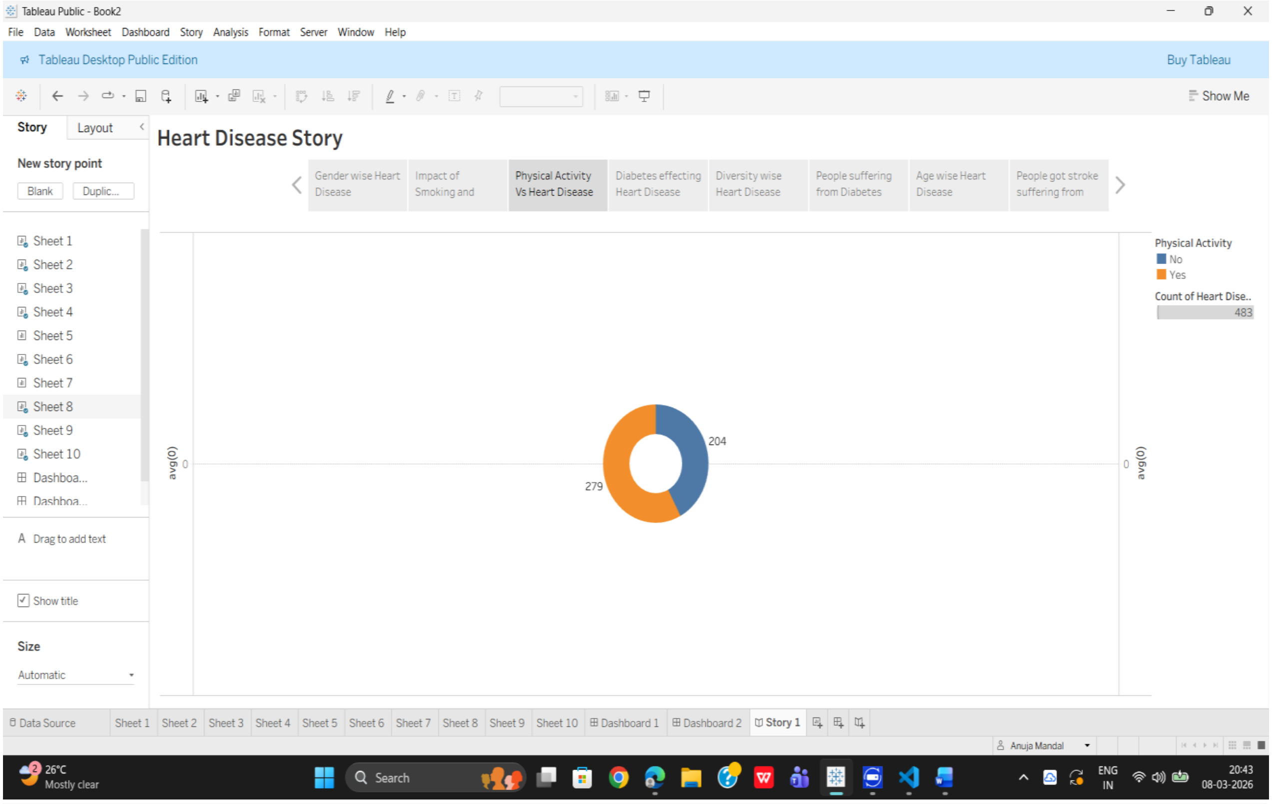
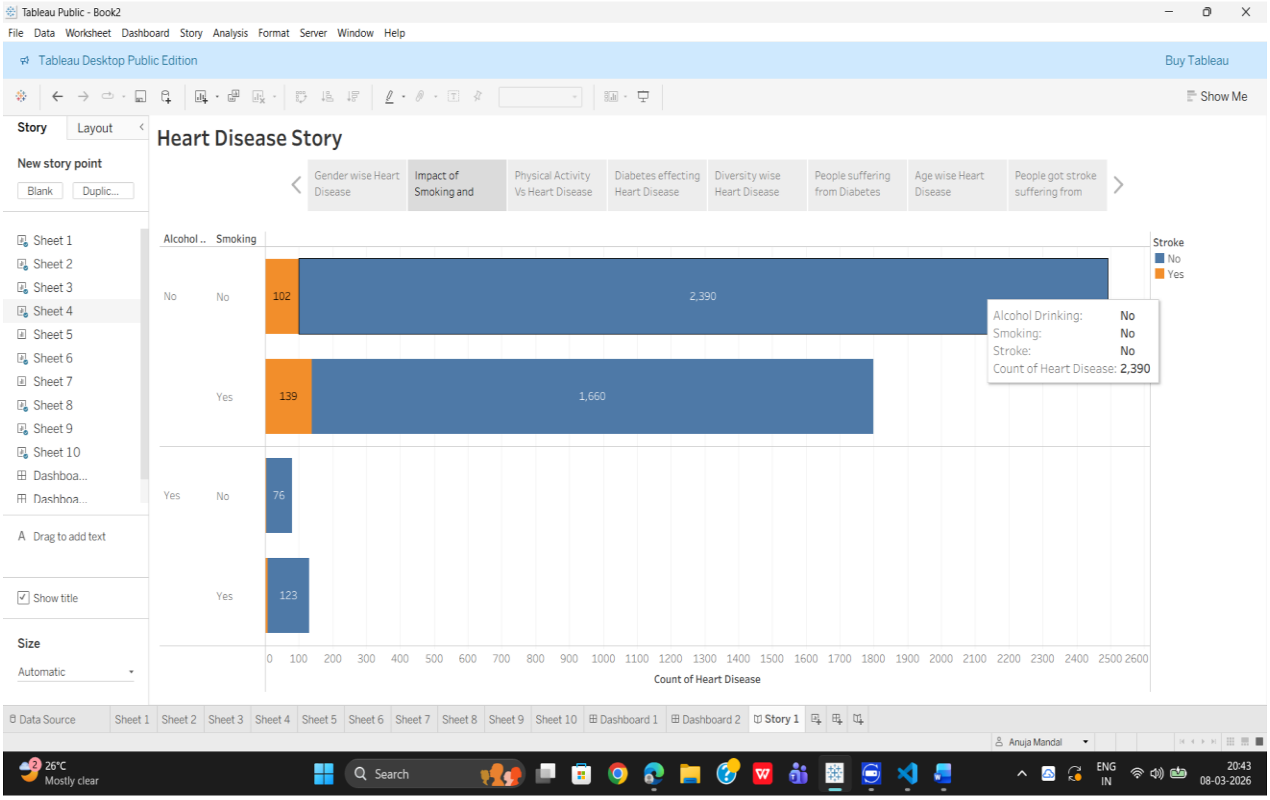
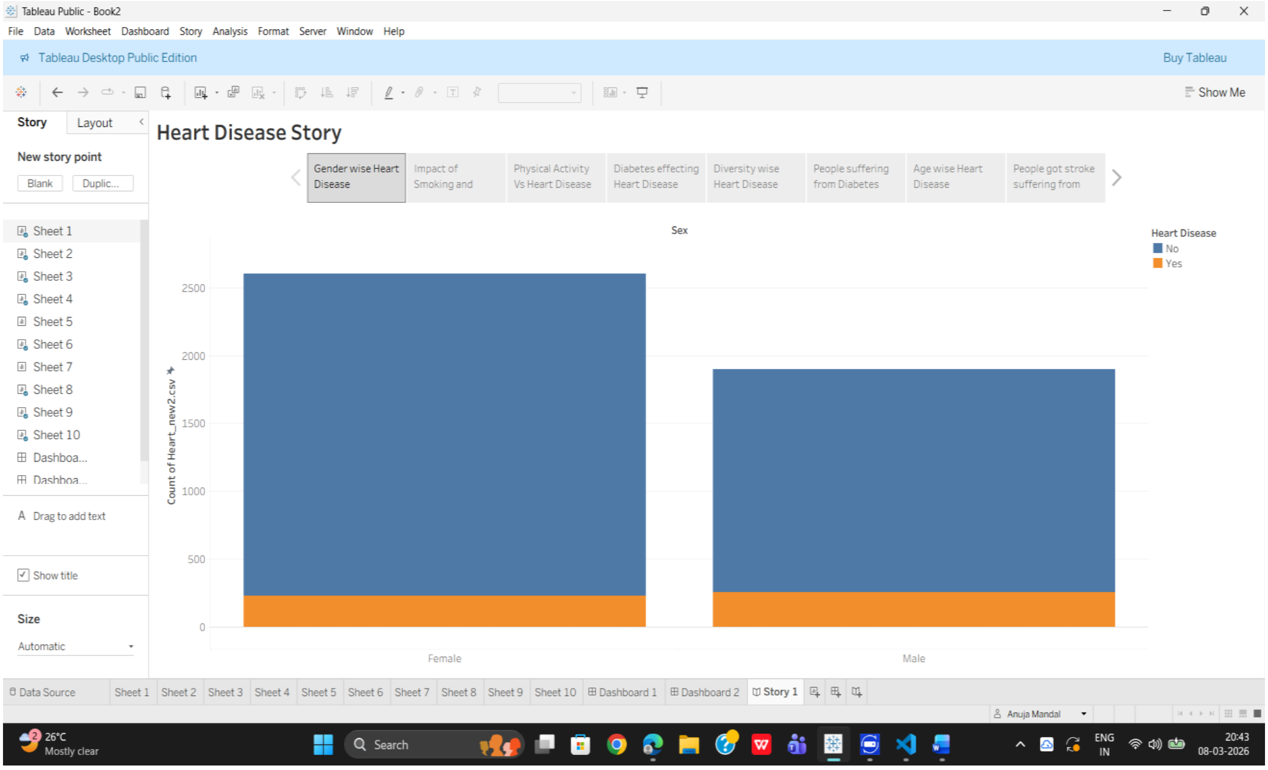


Tableau Public - Book2
Gender Vs Heart Disease
Sex
Count of Heart_new2.csv
Female
Male
Heart Disease
No
Yes
4 marks 1 row by 2 columns SUM of CNT(Heart_new2.csv): 4,500

- Tableau Story



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File Data Worksheet Dashboard Story Analysis Format Server Window Help
Tableau Desktop Public Edition
Buy Tableau
Show Me
Story
Layout
New story point
Blank
Duplic...
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Sheet 9
Sheet 10
Dashboa...
Dashboa...
Drag to add text
Show title
Size
Automatic
Heart Disease Story
Gender wise Heart Disease
Impact of Smoking and
Physical Activity Vs Heart Disease
Diabetes effecting Heart Disease
Diversity wise Heart Disease
People suffering from Diabetes
Age wise Heart Disease
People got stroke suffering from
Sex
Heart Disease
No
Yes
Count of Heart_new2.csv
2500
2000
1500
1000
500
0
Female
Male
Data Source
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Sheet 9
Sheet 10
Dashboard 1
Dashboard 2
Story 1
Anuja Mandal
26°C
Mostly clear
Search
ENG
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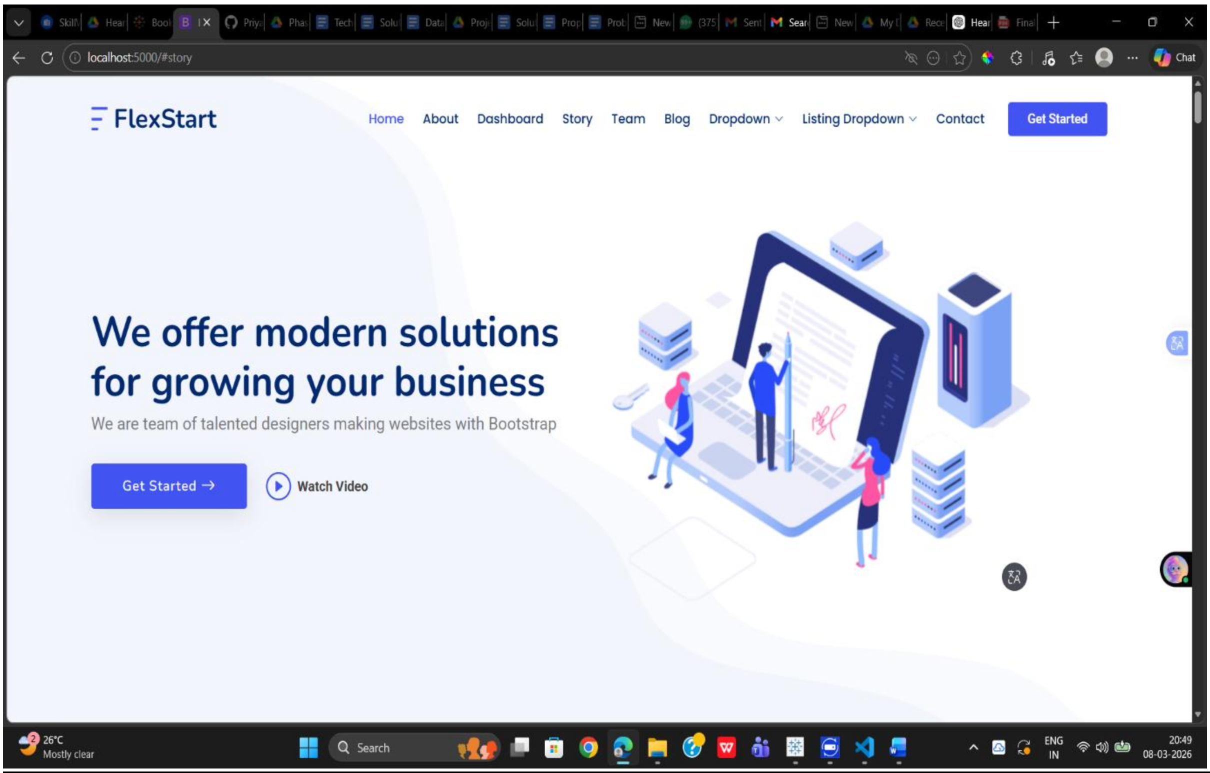
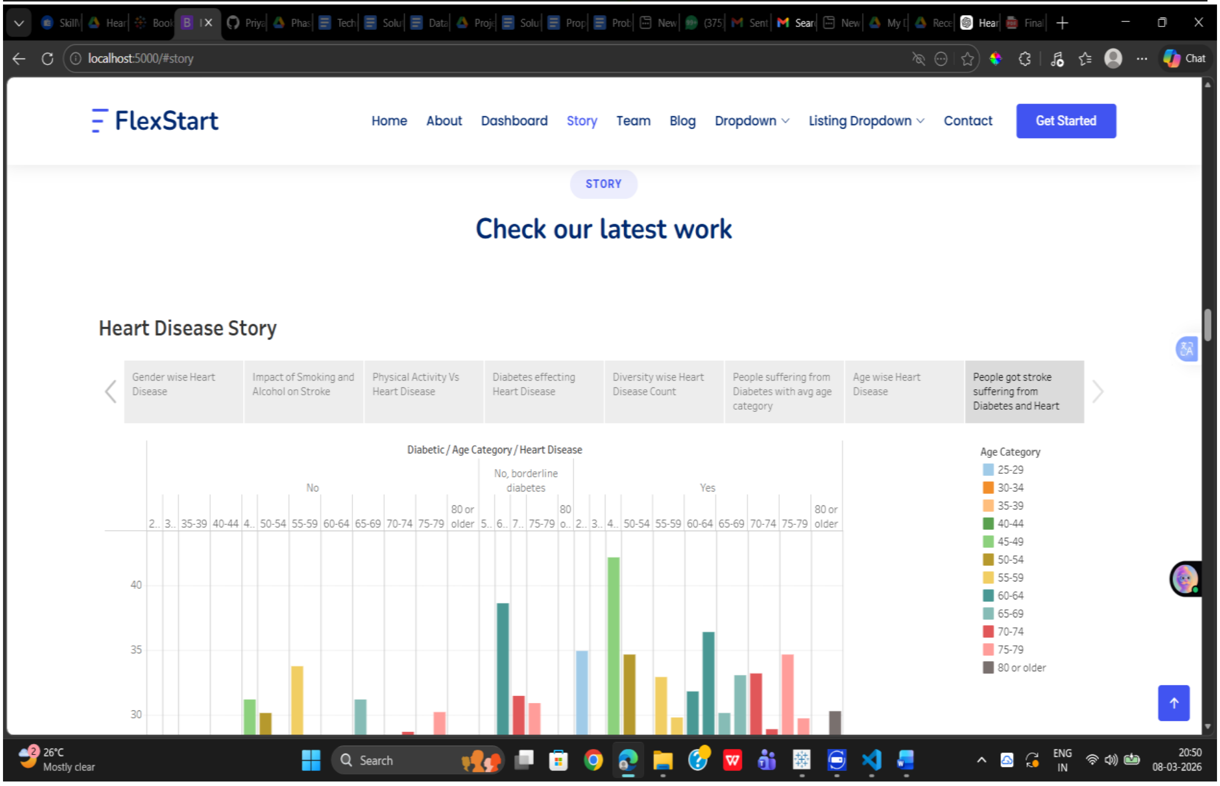
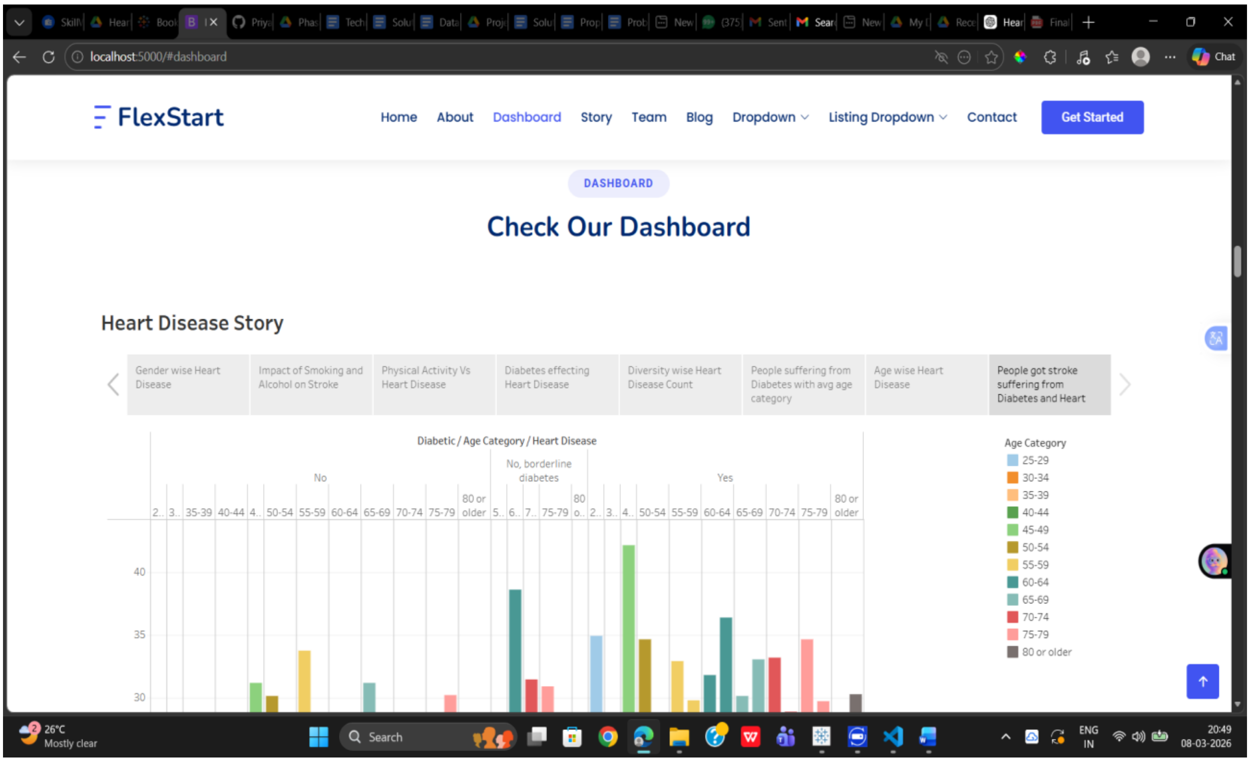
20:43
08-03-2026

- Flask Web Interface







These outputs demonstrate how the system visualizes heart disease data and provides meaningful insights.

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8. ADVANTAGES & DISADVANTAGES

Advantages

- Easy visualization of healthcare data
- Interactive dashboards
- Helps identify high-risk patient groups
- Supports healthcare decision making

Disadvantages

- Depends on the availability of accurate datasets
- Requires basic knowledge of Tableau
- Limited predictive capability without machine learning models

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9. CONCLUSION

The Heart Disease Analysis project successfully demonstrates how Tableau dashboards can be used to analyze healthcare data effectively.

By transforming raw datasets into visual insights, the system helps users understand the relationships between different health indicators and heart disease risk factors. This approach improves data interpretation and supports better healthcare decisions.

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10. FUTURE SCOPE

The project can be further improved by:

- Integrating machine learning models for heart disease prediction
- Adding real-time healthcare data integration
- Developing a mobile application for easier access
- Expanding the system to analyze other diseases

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11. APPENDIX

Dataset Link

Heart_new2.csv - Google Drive

GitHub Repository

https://github.com/Priyadeb-Barman/Heart-Disease-Analysis.git

Project Demo

https://drive.google.com/file/d/1of_j4l1bfe-qii3gBLFkMxqrbULt9M8w/view?usp=drive_link